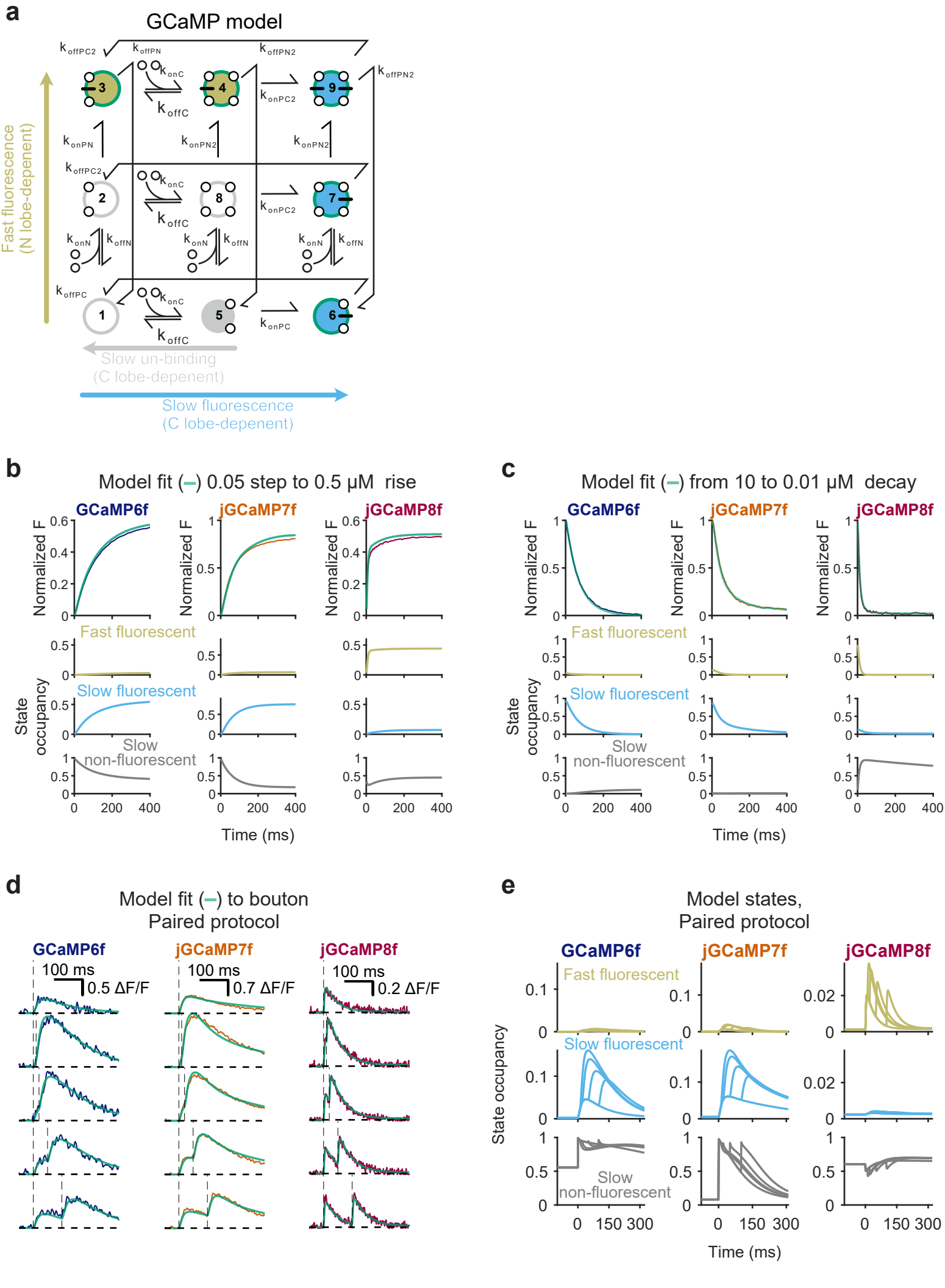

a
GCaMP model
Fast fluorescence
(N lobe-depenent)
Slow un-binding
(C lobe-depenent)
Slow fluorescence
(C lobe-depenent)
b
Model fit (–) 0.05 step to 0.5 μM rise
GCaMP6f
jGCaMP7f
jGCaMP8f
Normalized F
Fast fluorescent
Slow fluorescent
Slow
non-fluorescent
State occupancy
Time (ms)
c
Model fit (–) from 10 to 0.01 μM decay
GCaMP6f
jGCaMP7f
jGCaMP8f
Normalized F
Fast fluorescent
Slow fluorescent
Slow
non-fluorescent
State occupancy
Time (ms)
d
Model fit (–) to bouton
Paired protocol
GCaMP6f
jGCaMP7f
jGCaMP8f
100 ms
0.5 ΔF/F
100 ms
0.7 ΔF/F
100 ms
0.2 ΔF/F
e
Model states,
Paired protocol
GCaMP6f
jGCaMP7f
jGCaMP8f
Fast fluorescent
Slow fluorescent
Slow
non-fluorescent
State occupancy
Time (ms)